Milestone 3 | M02

Machine Delta Change Data Feed

SUBMISSION TEMPLATE

Difficulty	Marks	Tasks	Working budget
Medium	20	5	35 minutes

Trainee: _____Soumiya Ayyappan_____ Candidate ID: _____10877547_____ Catalog: _____

Objective

Track one update, one deletion and one insertion through Delta Change Data Feed.

Evidence policy

There is NO fixed one-task/one-screenshot limit. Attach clear and readable images for every assessed criterion. If code AND results are required, show BOTH: separate Code_A/Code_B and Result_A/Result_B slots may be used. When code does not fit, capture its consecutive portions in two or three screenshots; never shrink or stitch screenshots until unreadable. For UI-based tasks, show the exact configuration and saved output. Preserve outputs: do not repeat MERGE/UPDATE/DELETE or restart pipelines merely to photograph them. The evaluator scores only submitted visible evidence. A workspace URL is useful for reference, but does not replace evidence.

Task requirements and evidence holders

Task 1. Create CDF-enabled table | 4 marks

1.1 Create question EXTERNAL volume, upload both CSVs and read the baseline snapshot.

1.2 Create managed Delta table ms3_s1_m02_machines and enable Change Data Feed before any DML.

1.3 Display four baseline records and note the actual baseline table version.

Expected output

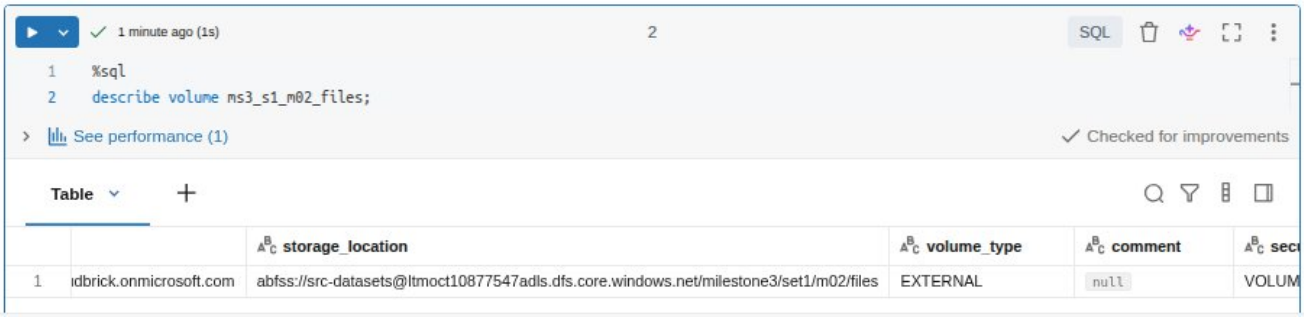
Baseline M101–M104; 4 rows; CDF enabled.

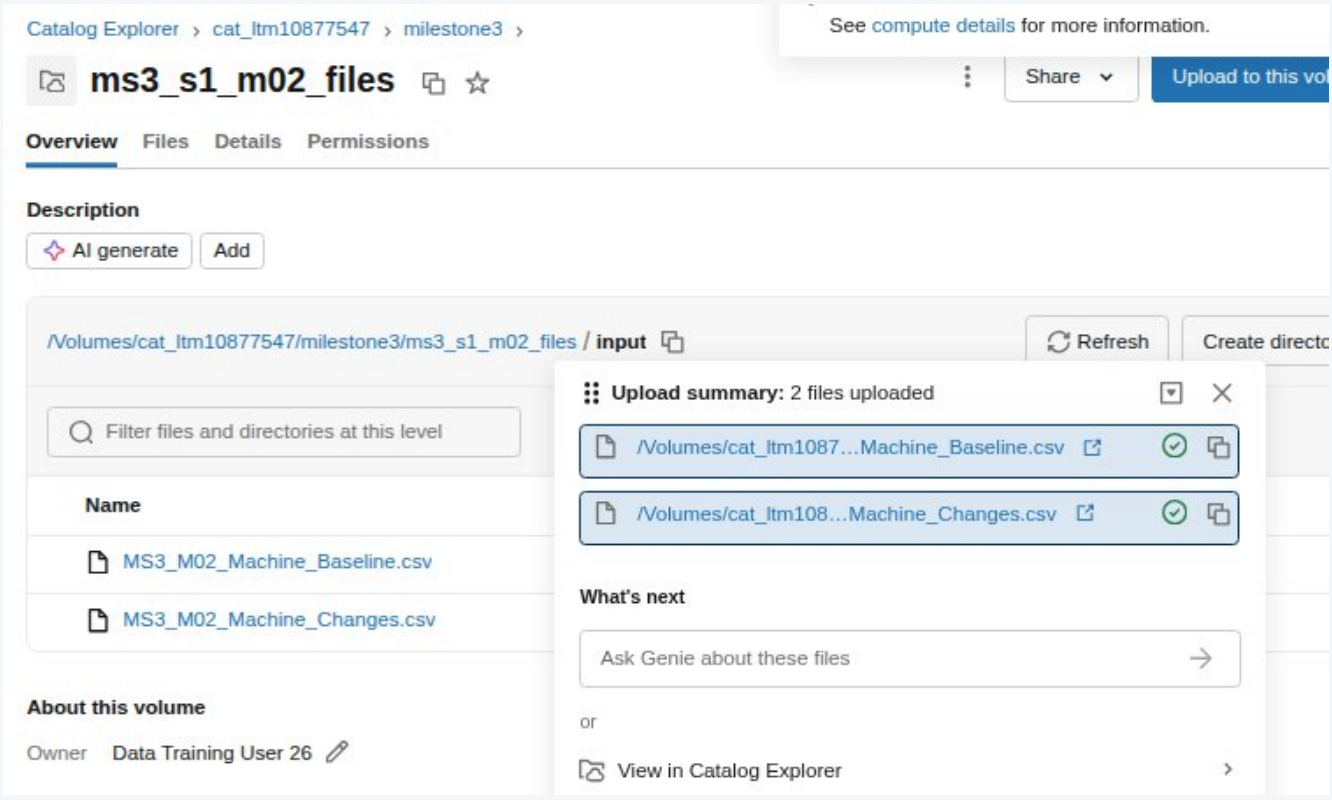
Required screenshots (code / result / UI)

A. [UI] External volume and both files. Suggested name: MS3_S1_M02_T01_UI_A.png. Additional numbered screenshots are allowed when necessary.

PASTE UI SCREENSHOT A BELOW:







B. [Code] Delta target creation plus CDF setting. Suggested name: MS3_S1_M02_T01_Code_B.png. Additional numbered screenshots are allowed when necessary.

PASTE CODE SCREENSHOT B BELOW:

09:45 AM (1s) 3

```
from pyspark.sql.functions import *
from pyspark.sql.types import *
schema = StructType([
    StructField("machine_id", StringType(), True),
    StructField("site", StringType(), True),
    StructField("temperature", IntegerType(), True),
    StructField("machine_status", StringType(), True)
])
path="/Volumes/cat_ltm10877547/milestone3/ms3_s1_m02_files/input/MS3_M02_Machine_Baseline.csv"
df=spark.read.csv(path,header=True,schema=schema)

```

df: pyspark.sql.connect.dataframe.DataFrame = [machine_id: string, site: string ... 2 more fields]

4 minutes ago (4s) 4

```
empty_df=spark.createDataFrame([],schema)
empty_df.write.format("delta").mode("overwrite").option("delta.EnableChangeDataFeed","true").saveAsTable("cat_ltm10877547.milestone3.ms3_s1_m02_machines")
df.write.format("delta").mode("append").option("mergeSchema","true").saveAsTable("cat_ltm10877547.milestone3.ms3_s1_m02_machines")

```

See performance (2) Checked for improvements

empty_df: pyspark.sql.connect.dataframe.DataFrame = [machine_id: string, site: string ... 2 more fields]

3 minutes ago (1s) 5 Python

```
baseline_df=spark.sql("DESCRIBE HISTORY cat_ltm10877547.milestone3.ms3_s1_m02_machines").first()["version"]
print(baseline_df)
```

See performance (1) Checked for improvements

3 minutes ago (1s)

```
display(df)
```

See performance (1)

C. [Result] Baseline 4 records and actual version. Suggested name: MS3_S1_M02_T01_Result_C.png. Additional numbered screenshots are allowed when necessary.

PASTE RESULT SCREENSHOT C BELOW:

```
print(baseline_df)
```

See performance (1)

1

Table +

	machine_id	site	temperature	machine_status
1	M101	SITE_A	65	RUNNING
2	M102	SITE_B	72	RUNNING
3	M103	SITE_A	68	RUNNING
4	M104	SITE_C	60	RUNNING

4 rows | 0.729s runtime

Extra screenshot(s) for this task: _____ Workspace reference: ____/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/medfinal2_____

Task 2. Classify changes | 3 marks

2.1 Read changes file and identify action for each machine_id.

2.2 Check unique machine IDs and correct type conversion.

Expected output

M102 UPDATE; M104 DELETE; M105 INSERT; exactly 3 change instructions.

Required screenshots (code / result / UI)

A. [Code] Typed change source and action selection. Suggested name: MS3_S1_M02_T02_Code_A.png. Additional numbered screenshots are allowed when necessary.

PASTE CODE SCREENSHOT A BELOW:

```python
changes_path = "/Volumes/cat_ltm10877547/milestone3/ms3_s1_m02_files/input/MS3_M02_Machine_Changes.csv"

changes_schema = StructType([
    StructField("machine_id", StringType(), True),
    StructField("site", StringType(), True),
    StructField("temperature", IntegerType(), True),
    StructField("machine_status", StringType(), True),
    StructField("action", StringType(), True)
])

changes_df = spark.read.csv(changes_path, header=True, schema=changes_schema)


print("Unique machine IDs and their actions:")
changes_df.select("machine_id", "action").distinct().show(truncate=False)

print(f"Total change instructions: {changes_df.count()}")

display(changes_df)
```

B. [Result] Three IDs/actions. Suggested name: MS3_S1_M02_T02_Result_B.png. Additional numbered screenshots are allowed when necessary.

PASTE RESULT SCREENSHOT B BELOW:

changes_df: pyspark.sql.connect.dataframe.DataFrame = [machine_id: string, site: string ... 3 more fields]

```
Unique machine IDs and their actions:
+----------+------+
|machine_id|action|
+----------+------+
|M102      |UPDATE|
|M104      |DELETE|
|M105      |INSERT|
+----------+------+

Total change instructions: 3
```

Table

	machine_id	site	temperature	machine_status	action
1	M102	SITE_B	85	ALERT	UPDATE
2	M104	SITE_C	60	RUNNING	DELETE
3	M105	SITE_B	70	RUNNING	INSERT

Extra screenshot(s) for this task: _____ Workspace reference: _____

Task 3. Apply three changes | 4 marks

3.1 Apply UPDATE to M102 temperature and status; DELETE M104; INSERT M105.

3.2 Execute the mutation sequence once; capture completed results without re-running.

Expected output

M102 becomes 85/ALERT; M104 absent; M105 70/RUNNING; final count 4.

Required screenshots (code / result / UI)

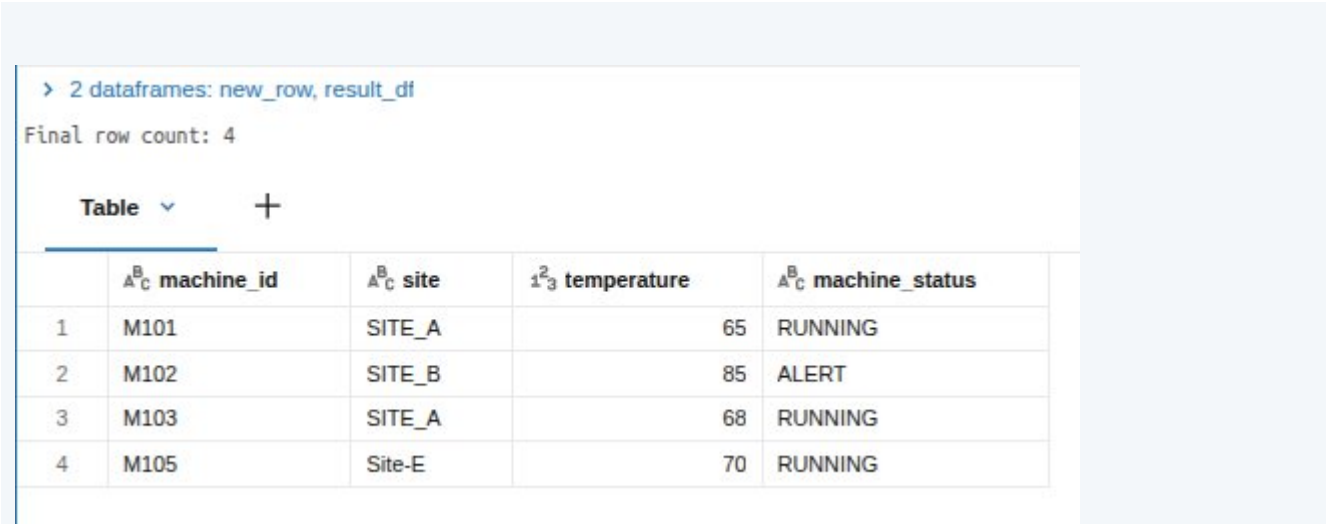
A. [Code] All three targeted DML operations and keys; split long code if necessary. Suggested name: MS3_S1_M02_T03_Code_A.png. Additional numbered screenshots are allowed when necessary.

PASTE CODE SCREENSHOT A BELOW:



B. [Result] Current target state and 4 rows. Suggested name: MS3_S1_M02_T03_Result_B.png. Additional numbered screenshots are allowed when necessary.

PASTE RESULT SCREENSHOT B BELOW:



	machine_id	site	temperature	machine_status
1	M101	SITE_A	65	RUNNING
2	M102	SITE_B	85	ALERT
3	M103	SITE_A	68	RUNNING
4	M105	Site-E	70	RUNNING

Extra screenshot(s) for this task: _____ Workspace reference: _____

Task 4. Read CDF event types | 4 marks

4.1 Read CDF for a version range beginning after the saved baseline version and ending at the actual post-change version.

4.2 Show machine_id, _change_type, _commit_version and relevant changed values.

Expected output

One update_preimage and one update_postimage for M102; one delete M104; one insert M105: four event rows.

Required screenshots (code / result / UI)

A. [Code] Correct CDF version range and metadata field selection. Suggested name: MS3_S1_M02_T04_Code_A.png. Additional numbered screenshots are allowed when necessary.

PASTE CODE SCREENSHOT A BELOW:

```
baseline_version = 1
post_change_version = spark.sql(
    "DESCRIBE HISTORY cat_ltm10877547.milestone3.ms3_s1_m02_machines"
).orderBy(col("version").desc()).first()["version"]

cdf_df = spark.read.format("delta") \
    .option("readChangeFeed", "true") \
    .option("startingVersion", baseline_version + 1) \
    .option("endingVersion", post_change_version) \
    .table("cat_ltm10877547.milestone3.ms3_s1_m02_machines")


cdf_df.select(
    "machine_id", "_change_type", "_commit_version",
    "site", "temperature", "machine_status"
).orderBy("machine_id", "_change_type").show(truncate=False)
```

B. [Result] All four event rows and commit versions. Suggested name: MS3_S1_M02_T04_Result_B.png. Additional numbered screenshots are allowed when necessary.

PASTE RESULT SCREENSHOT B BELOW:

```
+----------+----------------+---------------+------+-----------+--------------+
|machine_id|_change_type    |_commit_version|site  |temperature|machine_status|
+----------+----------------+---------------+------+-----------+--------------+
|M102      |update_postimage|2              |SITE_B|85         |ALERT         |
|M102      |update_preimage |2              |SITE_B|72         |RUNNING       |
|M104      |delete          |3              |SITE_C|60         |RUNNING       |
|M105      |insert          |5              |Site-E|70         |RUNNING       |
+----------+----------------+---------------+------+-----------+--------------+
```

Extra screenshot(s) for this task: _____ Workspace reference: _____

Task 5. Reconcile final state | 5 marks

5.1 Count current machine rows and CDF events grouped by _change_type.

5.2 Compare M102 before/after temperature and current saved rows.

Expected output

Current count 4; one each of insert, delete, update_preimage, update_postimage; M102 72→85.

Required screenshots (code / result / UI)

A. [Code] CDF group counts and saved-table validation. Suggested name: MS3_S1_M02_T05_Code_A.png. Additional numbered screenshots are allowed when necessary.

PASTE CODE SCREENSHOT A BELOW:

```python
table_name = "cat_ltm10877547.milestone3.ms3_s1_m02_machines"

# 5.1 Current machine row count and CDF events grouped by _change_type
current_df = spark.table(table_name)
print(f"Current machine rows: {current_df.count()}")

print("CDF events by _change_type:")
cdf_df.groupBy("_change_type").count().orderBy("_change_type").show(truncate=False)

# 5.2 M102 before/after temperature (baseline version vs current) and current saved rows
before_m102 = spark.read.option("versionAsOf", baseline_version).table(table_name) \
    .filter(col("machine_id") == "M102").select(col("temperature").alias("before_temperature"), col("machine_status").alias("before_status"))
after_m102 = current_df.filter(col("machine_id") == "M102") \
    .select(col("temperature").alias("after_temperature"), col("machine_status").alias("after_status"))

print("M102 before/after:")
before_m102.crossJoin(after_m102).show(truncate=False)

print("Current saved rows:")
current_df.orderBy("machine_id").show(truncate=False)
```

B. [Result] Four event counts, M102 before/after and current 4. Suggested name: MS3_S1_M02_T05_Result_B.png. Additional numbered screenshots are allowed when necessary.

PASTE RESULT SCREENSHOT B BELOW:

```
Current machine rows: 4
CDF events by _change_type:
+----------------+-----+
|_change_type    |count|
+----------------+-----+
|delete          |1    |
|insert          |1    |
|update_postimage|1    |
|update_preimage |1    |
+----------------+-----+

M102 before/after:
+------------------+-------------+-----------------+------------+
|before_temperature|before_status|after_temperature|after_status|
+------------------+-------------+-----------------+------------+
|72                |RUNNING      |85               |ALERT       |
+------------------+-------------+-----------------+------------+
```

```
Current saved rows:
+----------+------+-----------+--------------+
|machine_id|site  |temperature|machine_status|
+----------+------+-----------+--------------+
|M101      |SITE_A|65         |RUNNING       |
|M102      |SITE_B|85         |ALERT         |
|M103      |SITE_A|68         |RUNNING       |
|M105      |Site-E|70         |RUNNING       |
+----------+------+-----------+--------------+
```

Extra screenshot(s) for this task: _____ Workspace reference: ___/Workspace/Users/datauser26@financecloudbrick.onmicrosoft.com/medfinal2_____

Expected final results

Task 1: Baseline M101–M104; 4 rows; CDF enabled.

Task 2: M102 UPDATE; M104 DELETE; M105 INSERT; exactly 3 change instructions.

Task 3: M102 becomes 85/ALERT; M104 absent; M105 70/RUNNING; final count 4.

Task 4: One update_preimage and one update_postimage for M102; one delete M104; one insert M105: four event rows.

Task 5: Current count 4; one each of insert, delete, update_preimage, update_postimage; M102 72→85.

Evaluation rubric / evidence map

Task	Criterion	Marks	Evidence
1	Correct source and target	2	UI+Code
1	CDF enabled and baseline version recorded	2	Code+Result
2	Correct typed change source	2	Code
2	Correct 3-event classification	1	Result
3	Correct key-based update/delete/insert	3	Code

3	Correct current saved rows	1	Result
4	Correct CDF read/version boundaries	2	Code
4	Correct four event types and IDs	2	Result
5	Correct reconciliation logic	3	Code
5	Correct final and event totals	2	Result

Submission requirements

Submit this question's matching template with readable taskwise screenshots. Screenshots may exceed the suggested slot count; both code and output must be present when rubric requires both. Keep saved objects and version history available; use the actual version identifiers. No executable solution code or validation scripts are supplied in this document. There are no hidden rubric criteria.